6 Ablation Study

Impact of Problem-Distiller As illustrated in Figure 7, when the problem-distiller is disabled, both Llama3-70B and GPT-4 experience a certain degree of accuracy decline. More complex problems, such as Game of 24 and Checkmate-in-One, show a more significant accuracy reduction, whereas relatively simpler problems like word list sorting and MGSM exhibit smaller decreases. This is because LLMs can more easily extract key information in simpler tasks, making the impact of the problem-distiller less noticeable. In contrast, extracting key information and potential constraints in complex problems is more challenging, making the role of our problem-distiller more prominent, thereby explaining the differences depicted in the figure.

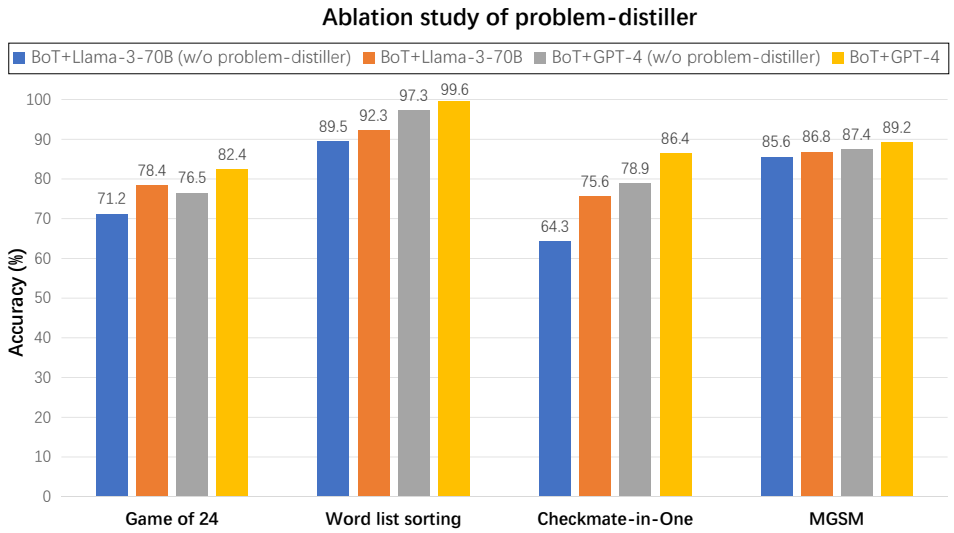

Figure 7: We conduct ablation study on problem-distiller across four benchmarks, employing Llama3-70B and GPT-4 as the base models.

Impact of Meta-Buffer As illustrated in Figure 8, when the meta-buffer is disabled, both Llama3-70B and GPT-4 models exhibit a noticeable decline in performance, particularly in benchmarks requiring complex reasoning, such as Game of 24 and Checkmate-in-One. This further underscores the superiority of our meta-buffer in addressing complex problems.

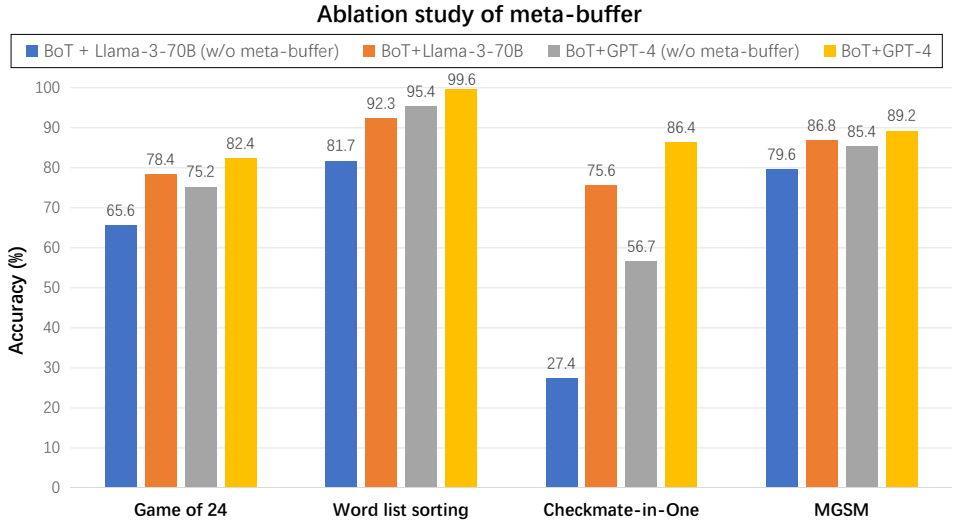

Figure 8: We conduct ablation study on meta-buffer across four benchmarks, employing Llama3-70B and GPT-4 as the base models.

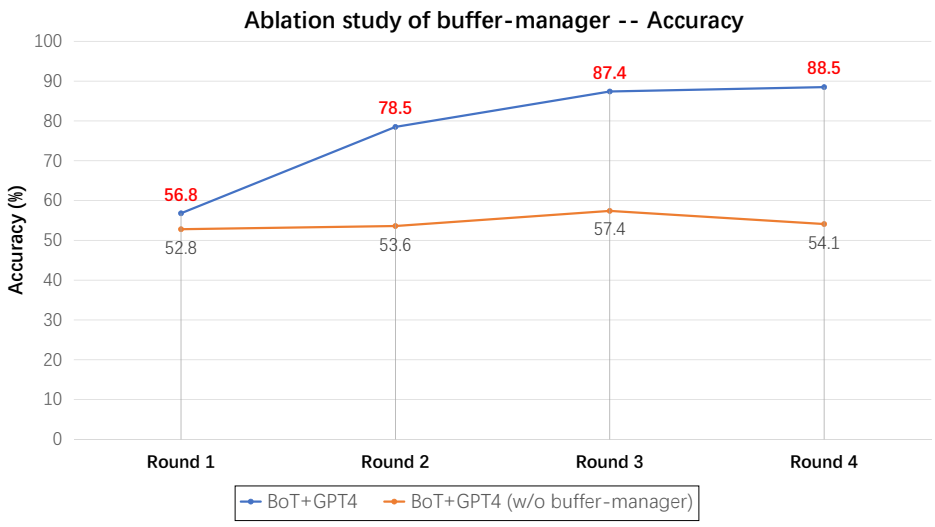

Figure 9: We conduct ablation study on buffer-manager regarding reasoning accuracy across four tasks, employing Llama3-70B and GPT-4 as the base models.

Impact of Buffer-Manager In this ablation study, we divide the entire process into four rounds. In each round, we randomly sample 50 questions from each benchmark and conduct reasoning. In the subsequent round, we continue to randomly sample another 50 questions from each benchmark. As depicted in Figure 9, with the increase of the number of rounds, the model with the buffer-manager continually expands the meta-buffer while also utilizing the thought-templates obtained from previously solved problems to help addressing subsequent similar problems. Therefore, we can observe that the accuracy of BoT steadily improves with each round. In contrast, the model without the buffer-manager fails to exhibit an upward trend. Additionally, we have also measured the reasoning time as depicted in Figure 10. when the number of rounds increases, the model with the buffer-manager will experience a continual improvement in reasoning efficiency. This is because, with the continual expansion of the meta-buffer, the likelihood of retrieving suitable thought-templates also increases. Consequently, models can avoid constructing reasoning structures from scratch, thereby enhancing the inference efficiency accordingly.

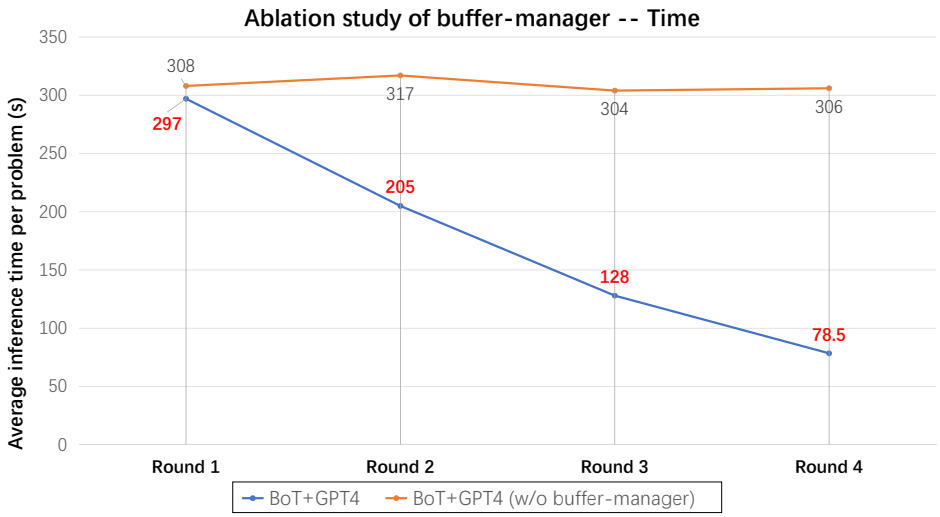

Figure 10: We conduct ablation study on buffer-manager regarding reasoning efficiency across four tasks, employing Llama3-70B and GPT-4 as the base models.

7 Discussion

Limitations and Future Directions Despite our method's significant improvement in accuracy while maintaining reasoning efficiency and robustness, our method's enhancements are limited when addressing problems requiring human-like creativity, as this issue often does not rely on a specific thought-template. Besides, if our BoT initializes the meta-buffer with a weaker model, the quality of the derived thought-templates may be suboptimal due to the weaker model's limited reasoning ability and instruction-following capability. Overall, our BoT brings out a set of future directions: 1. integrating external resources with BoT to build a open-domain system like agent models [54, 55]. 2. making the distillation of thought-templates optimizable, which may significantly enhance their template qualities for more complex tasks.

Conclusion In this work, we introduce Buffer of Thoughts, a novel beffered reasoning framework that employs LLMs to utilize pre-accumulated experiences and methodologies from prior tasks as thought-templates stored within a meta-buffer. We further design buffer-manager to continuously refine the problem-solving processes and dynamically distill thought-templates, thereby progressively raising the LLM's reasoning capacity. Our BoT demonstrates SOTA performance on 10 challenging tasks, and offers promising prospects for future research and application.

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